

# Method: misspecified models & core code

**Setup (Sec. IV).** True DGP:  $P$  signals,  $c = P/T$ . We see the first  $P_1$ ,  $q = P_1/P$ ; model complexity  $cq$ .

**Prop. 6** (ridgeless,  $\Psi = \psi I$ ):

$$\mathcal{E}(0; cq; q) = b_* \psi \min\{q, c^{-1}\}$$

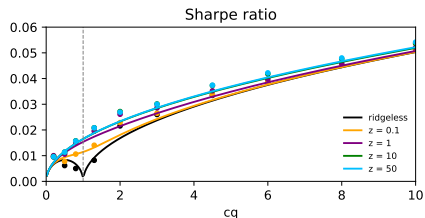
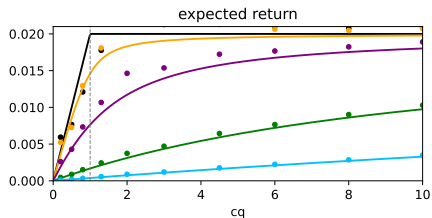
flat for  $cq > 1$ : approximation gain = min-norm bias.

**Theorem 1.** With optimal  $z_*$ :  $SR$ ,  $R^2$  strictly increasing & concave in  $q$ .

Double ascent  $\rightarrow$  permanent ascent.

**Core module** (`voc_core.py`):

- ▶ RFF:  $[\sin(\gamma\omega'G), \cos(\gamma\omega'G)]$ ,  
 $\omega \sim N(0, I)$ ,  $\gamma = 2$
- ▶  $\hat{\beta} = S'(zT I_T + SS')^{-1}R$ ;  $z = 0 \Rightarrow$   
pseudo-inverse
- ▶  $P = 12,000$ , 7  $z$ , 1,079 windows:  $\approx 1$  s



$\Psi = I$ ,  $b_* = 0.2$ ,  $c = 10$ . Lines: closed form (Prop. 5); dots: Monte Carlo,  $T = 200$ , 20 sims. Cf. Figs. 5–6.

## Next steps

1. **Real data:** Goyal–Welch predictors  $G_t$  (15)  $\rightarrow$  `rff()`  $\rightarrow$  rolling ridge
2. **Backtest (Figs. 7–8):**  $T = 12$ ,  $P$  up to 12,000,  $\log_{10} z \in \{-3, \dots, 3\}$   
10–30 seeds instead of the paper's 1,000, reported openly
3. **Table I, linear rows:** same engine with  $P = 15$ ; ridgeless via pseudo-inverse
4. **Robustness:**  $T \in \{60, 120\}$ ,  $\gamma \in \{0.5, 1\}$
5. **Critique check:** at  $T = 12$  with persistent predictors the model is close to time-series momentum (authors, p. 499)